

Below is the complete MySQL 8.0 DDL script for all 15 tables in the **Inventory Management System** project. It includes:

- Database Creation
 - 15 Tables
 - Primary Keys
 - Foreign Keys
 - Unique Constraints
 - Check Constraints
 - Default Values
 - Indexes
-

Database Creation

```
1 CREATE DATABASE IF NOT EXISTS InventoryManagementDB;
2
3 USE InventoryManagementDB;
```

1. Roles

```
1 CREATE TABLE Roles
2 (
3     RoleID INT AUTO_INCREMENT PRIMARY KEY,
4     RoleName VARCHAR(50) NOT NULL UNIQUE,
5     Description VARCHAR(255)
6 );
```

2. Users

```
1 CREATE TABLE Users
2 (
3     UserID INT AUTO_INCREMENT PRIMARY KEY,
4     RoleID INT NOT NULL,
5     Username VARCHAR(100) NOT NULL,
6     Email VARCHAR(100) UNIQUE,
```

```
7     Phone VARCHAR(20),
8     CreatedDate DATETIME DEFAULT CURRENT_TIMESTAMP,
9
10    CONSTRAINT FK_Users_Roles
11    FOREIGN KEY (RoleID)
12    REFERENCES Roles(RoleID)
13 );
```

3. Suppliers

```
1 CREATE TABLE Suppliers
2 (
3     SupplierID INT AUTO_INCREMENT PRIMARY KEY,
4     SupplierName VARCHAR(150) NOT NULL,
5     ContactPerson VARCHAR(100),
6     Phone VARCHAR(20),
7     Email VARCHAR(100),
8     City VARCHAR(100),
9     State VARCHAR(100),
10
11     Status ENUM('Active','Inactive')
12     DEFAULT 'Active'
13 );
```

4. Categories

```
1 CREATE TABLE Categories
2 (
3     CategoryID INT AUTO_INCREMENT PRIMARY KEY,
4     CategoryName VARCHAR(100) NOT NULL UNIQUE,
5     Description VARCHAR(255)
6 );
```

5. SubCategories

```
1 CREATE TABLE SubCategories
2 (
```

```
3     SubCategoryID INT AUTO_INCREMENT PRIMARY KEY,  
4  
5     CategoryID INT NOT NULL,  
6  
7     SubCategoryName VARCHAR(100) NOT NULL,  
8  
9     CONSTRAINT FK_SubCategories_Categories  
10    FOREIGN KEY (CategoryID)  
11    REFERENCES Categories(CategoryID)  
12 );
```

6. Products

```
1 CREATE TABLE Products  
2 (  
3     ProductID INT AUTO_INCREMENT PRIMARY KEY,  
4  
5     SubCategoryID INT NOT NULL,  
6  
7     ProductName VARCHAR(200) NOT NULL,  
8  
9     SKU VARCHAR(50) UNIQUE NOT NULL,  
10  
11     UnitPrice DECIMAL(10,2) NOT NULL,  
12  
13     ReorderLevel INT DEFAULT 10,  
14  
15     Status ENUM('Active','Inactive')  
16     DEFAULT 'Active',  
17  
18     CONSTRAINT FK_Products_SubCategories  
19     FOREIGN KEY (SubCategoryID)  
20     REFERENCES SubCategories(SubCategoryID)  
21 );
```

7. Warehouses

```
1 CREATE TABLE Warehouses
```

```
2 (
3     WarehouseID INT AUTO_INCREMENT PRIMARY KEY,
4
5     WarehouseName VARCHAR(150) NOT NULL,
6
7     City VARCHAR(100),
8
9     State VARCHAR(100),
10
11     Capacity INT NOT NULL
12 );
```

8. Customers

```
1 CREATE TABLE Customers
2 (
3     CustomerID INT AUTO_INCREMENT PRIMARY KEY,
4
5     CustomerName VARCHAR(150) NOT NULL,
6
7     Phone VARCHAR(20),
8
9     Email VARCHAR(100),
10
11     City VARCHAR(100),
12
13     CustomerType ENUM
14     (
15         'Retail',
16         'Wholesale',
17         'Distributor'
18     ) DEFAULT 'Retail'
19 );
```

9. PurchaseOrders

```
1 CREATE TABLE PurchaseOrders
2 (
```

```

3      PurchaseOrderID INT AUTO_INCREMENT PRIMARY KEY,
4
5      SupplierID INT NOT NULL,
6
7      OrderDate DATE NOT NULL,
8
9      OrderStatus ENUM
10     (
11         'Pending',
12         'Approved',
13         'Completed',
14         'Cancelled'
15     ) DEFAULT 'Pending',
16
17     TotalAmount DECIMAL(12,2)
18     DEFAULT 0,
19
20     CONSTRAINT FK_PO_Supplier
21     FOREIGN KEY (SupplierID)
22     REFERENCES Suppliers(SupplierID)
23 );

```

10. PurchaseOrderItems

```

1  CREATE TABLE PurchaseOrderItems
2  (
3      POItemID INT AUTO_INCREMENT PRIMARY KEY,
4
5      PurchaseOrderID INT NOT NULL,
6
7      ProductID INT NOT NULL,
8
9      Quantity INT NOT NULL,
10
11     UnitCost DECIMAL(10,2) NOT NULL,
12
13     LineAmount DECIMAL(12,2),
14
15     CONSTRAINT FK_POItems_PO

```

```
16 FOREIGN KEY (PurchaseOrderID)
17 REFERENCES PurchaseOrders(PurchaseOrderID),
18
19 CONSTRAINT FK_POItems_Product
20 FOREIGN KEY (ProductID)
21 REFERENCES Products(ProductID)
22 );
```

11. GoodsReceipts

```
1 CREATE TABLE GoodsReceipts
2 (
3     ReceiptID INT AUTO_INCREMENT PRIMARY KEY,
4
5     PurchaseOrderID INT NOT NULL,
6
7     WarehouseID INT NOT NULL,
8
9     ReceiptDate DATE NOT NULL,
10
11     ReceivedBy INT,
12
13     CONSTRAINT FK_GR_PO
14     FOREIGN KEY (PurchaseOrderID)
15     REFERENCES PurchaseOrders(PurchaseOrderID),
16
17     CONSTRAINT FK_GR_Warehouse
18     FOREIGN KEY (WarehouseID)
19     REFERENCES Warehouses(WarehouseID),
20
21     CONSTRAINT FK_GR_User
22     FOREIGN KEY (ReceivedBy)
23     REFERENCES Users(UserID)
24 );
```

12. SalesOrders

```
1 CREATE TABLE SalesOrders
```

```

2  (
3      SalesOrderID INT AUTO_INCREMENT PRIMARY KEY,
4
5      CustomerID INT NOT NULL,
6
7      OrderDate DATE NOT NULL,
8
9      OrderStatus ENUM
10     (
11         'Pending',
12         'Processing',
13         'Shipped',
14         'Delivered',
15         'Cancelled',
16         'Paid'
17     )
18     DEFAULT 'Pending',
19
20     TotalAmount DECIMAL(12,2)
21     DEFAULT 0,
22
23     CONSTRAINT FK_SO_Customer
24     FOREIGN KEY (CustomerID)
25     REFERENCES Customers(CustomerID)
26 );

```

13. SalesOrderItems

```

1  CREATE TABLE SalesOrderItems
2  (
3      SalesItemID INT AUTO_INCREMENT PRIMARY KEY,
4
5      SalesOrderID INT NOT NULL,
6
7      ProductID INT NOT NULL,
8
9      Quantity INT NOT NULL,
10
11     UnitPrice DECIMAL(10,2) NOT NULL,

```

```

12
13     LineAmount DECIMAL(12,2),
14
15     CONSTRAINT FK_SOItems_SO
16     FOREIGN KEY (SalesOrderID)
17     REFERENCES SalesOrders(SalesOrderID),
18
19     CONSTRAINT FK_SOItems_Product
20     FOREIGN KEY (ProductID)
21     REFERENCES Products(ProductID)
22 );

```

14. InventoryTransactions

```

1  CREATE TABLE InventoryTransactions
2  (
3      TransactionID INT AUTO_INCREMENT PRIMARY KEY,
4
5      ProductID INT NOT NULL,
6
7      WarehouseID INT NOT NULL,
8
9      TransactionType ENUM
10     (
11         'IN',
12         'OUT',
13         'ADJUSTMENT'
14     ) NOT NULL,
15
16     Quantity INT NOT NULL,
17
18     TransactionDate DATETIME
19     DEFAULT CURRENT_TIMESTAMP,
20
21     CONSTRAINT FK_IT_Product
22     FOREIGN KEY (ProductID)
23     REFERENCES Products(ProductID),
24
25     CONSTRAINT FK_IT_Warehouse

```



```
26     FOREIGN KEY (WarehouseID)
27     REFERENCES Warehouses(WarehouseID)
28 );
```

15. Payments

```
1  CREATE TABLE Payments
2  (
3      PaymentID INT AUTO_INCREMENT PRIMARY KEY,
4
5      SalesOrderID INT NOT NULL,
6
7      Amount DECIMAL(12,2) NOT NULL,
8
9      PaymentDate DATETIME
10     DEFAULT CURRENT_TIMESTAMP,
11
12     PaymentMode ENUM
13     (
14         'Cash',
15         'Card',
16         'UPI',
17         'Bank Transfer'
18     ),
19
20     CONSTRAINT FK_Payments_SO
21     FOREIGN KEY (SalesOrderID)
22     REFERENCES SalesOrders(SalesOrderID)
23 );
```

Indexes

```
1  CREATE INDEX IDX_Product_Name
2  ON Products(ProductName);
3
4  CREATE INDEX IDX_Product_SKU
5  ON Products(SKU);
6
```

```
7 CREATE INDEX IDX_Customer_Name
8 ON Customers(CustomerName);
9
10 CREATE INDEX IDX_PO_Date
11 ON PurchaseOrders(OrderDate);
12
13 CREATE INDEX IDX_SO_Date
14 ON SalesOrders(OrderDate);
15
16 CREATE INDEX IDX_IT_Product
17 ON InventoryTransactions(ProductID);
18
19 CREATE INDEX IDX_IT_Warehouse
20 ON InventoryTransactions(WarehouseID);
21
22 CREATE INDEX IDX_Payment_Date
23 ON Payments(PaymentDate);
```

Optional Audit Table (for Triggers)

```
1 CREATE TABLE ProductAudit
2 (
3     AuditID INT AUTO_INCREMENT PRIMARY KEY,
4
5     ProductID INT,
6
7     ActionType VARCHAR(20),
8
9     ActionDate DATETIME,
10
11     Remarks VARCHAR(255)
12 );
```

This DDL is MySQL 8.0 compatible.